The Power Law and Fan Effect Revisited

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Draft for Comment[1]

Abstract

In 1988, Tad Hogg, Bernardo Huberman and I proposed that the power law of practice need not be built into a learning mechanism: it could emerge from the dynamics of search in a problem space that learning makes more densely connected, and the same dynamics could make practice hurt, a problem-solving fan effect. The paper supported this with simulations of a simple "bit game" and with an analytic theory based on a random walk. Here I revisit that work by formalizing it in the Lean proof assistant, so that what the theory entails is checked by machine rather than read off curves. Most of the 1988 mathematics survives: the expected-steps formula is correct exactly as printed, and its drift and diffusion regimes hold, one of them in a sharper form. But two of the paper's conclusions do not follow from its own theory. With no decision procedure, the idealized model shows a fan effect in sparse graphs, which the paper's general statement rules out; and with a perfect decision procedure the "power-law decay" it derives is logarithmic, a curve with every qualitative signature of a power law over the range plotted, and provably not one over long ranges. In the bit game itself, exact results reproduce the pattern of the 1988 simulations. But the theory, started where the game starts, does not: at the size that was simulated it predicts steady improvement in every regime, so it does not explain the fan effect that the simulations showed. None of these findings needed new mathematics. They needed the theory written down precisely enough that its consequences could be checked.

1 Introduction

The power law of practice is one of the most robust regularities in the psychology of learning: the time to perform a task falls as a power of the number of times it has been performed [1–3]. In 1988 the leading accounts obtained it in quite different ways. ACT* obtained it from a strengthening mechanism that itself follows a power law [4]; Newell and Rosenbloom obtained it from the chunking of subgoals [3, 5]. Tad Hogg, Bernardo Huberman and I proposed a third account [6]. Problem solving is search through a problem-space graph; learning adds edges to the graph (new operators) and can improve the procedure that chooses among them. Neither change mentions a power law. Yet their interaction, under particular conditions, yields curves that look like one, and under other conditions yields the opposite: learning that makes performance worse. We called that the problem-solving fan effect, by analogy with the memory fan effect [7], and connected it to the utility problem in machine learning [8, 9].

That paper did its reasoning in two ways. It simulated a small task, the bit game, and it derived an analytic theory, a one-dimensional random walk whose expected length depends on how

[1] This is a working draft. I welcome corrections, constructive challenges, and especially proposals for collaboration. Email me: `jshrager@gmail.com`.

densely the graph is connected and how good the decision procedure is. Each regime of the walk was connected to a phenomenon: drift toward the goal to improvement with practice, a balance of branching against the shortening of paths to the fan effect. In this sense the 1988 paper already had the explanatory shape I have since been arguing for in CogLean[2] [10, 11]: mechanisms, together with stated conditions, entail phenomena, including conditions under which a phenomenon fails or reverses.

What it did not have was a way to check that its conclusions followed. The derivations were done by hand, the approximations were judged by eye, and the agreement between theory and simulation was read off log-log plots. So I have written the 1988 theory and the bit game in Lean [12, 13], where the definitions that run are the same objects that theorems are about, and every claim of the paper has been given a status: proved, sharpened, reinterpreted, or refuted (Table 3). The work is in a public repository [14]; every result below names the theorem that establishes it.

The outcome is a mixed verdict, which is the interesting kind. The mathematics of the walk is right. But the two conclusions the paper drew from it about practice do not follow: one fails in the paper's own idealized model outside a range the paper does not state, and the other names the wrong curve. And the theory and the game, which the paper treated as showing the same thing, turn out to come apart in a place nobody was looking.

2 The model

I restate the 1988 model briefly, as the formalization reads it; the full list of postulates, conditions and idealizations, with page references, is in `notes/POSTULATES.md`.

The bit game. A state is a vector of B bits. An operator is a pattern–result pair with don't-care positions: `?1?1?` $\rightarrow$ `?0?0?` applies to any state with 1s in the second and fourth positions and sets them to 0. Play begins with the $2B$ one-bit operators, which connect the states as a hypercube. A trial draws a start and a goal, and performance is the number of operators applied to reach the goal (`SHH.BitGame.apply`, `SHH.BitGame.initialOps`).

Learning. After each trial, the operator that solves that game in one step is added, with a don't-care wherever start and goal agree. For `01110` $\rightarrow$ `10101` it is `01?10` $\rightarrow$ `10?01` (`SHH.BitGame.chunk_paper`), which "connects two pairs of states": it applies to exactly two of the 32 states (`SHH.BitGame.chunk_paper_applies`). In general a learned operator takes its own start to its own goal (`SHH.BitGame.chunk_apply`).

The decision procedure. At a state, the applicable operators whose results are closest to the goal (by Hamming distance) are good, the rest bad; each bad one is discarded with probability r, and one operator is chosen at random from what remains. So $r = 1$ is a perfect procedure and $r = 0$ a random walk. The formalization computes the exact distribution over next states, with the probabilities of good and bad choices as binomial sums (`SHH.BitGame.stepDist`).

The theory. To analyze typical rather than particular problem spaces, the paper idealizes the graph as random, with n nodes and mean branching μ, and then as a tree: start and goal are about the diameter $D = \ln n / \ln \mu$ apart, and at each node one edge leads toward the goal and $\mu - 1$ away.

[2] CogLean is a library of cognitive theory in the Lean proof assistant, in which mechanisms, phenomena and laws are stated once and theorems relating them are machine-checked: `https://coglean.org`. This paper is a self-contained application of its approach to one theory.

The decision procedure leaves $(\mu - 1)(1 - r)$ effective bad choices, so a step goes toward the goal with probability

$$p_- = \frac{1}{1 + (\mu - 1)(1 - r)}. \tag{1}$$

The search becomes a walk on distance $0, \ldots, D$ from the goal (`SHH.GD.pOf`, `SHH.GD.DOf`). Everything in the theory depends on this idealization, and the formalization keeps it separate from the game, as an explicit postulate. That separation turns out to matter.

3 The theory, checked

The expected number of steps. The paper gives the expected number of steps s to reach the goal from distance D as its Eq. 2, obtained by "standard techniques" [15]. The formalization states the walk as a linear system for the expected steps from every distance (`SHH.Walk.Sys`), proves a closed form satisfies it (`SHH.Walk.term_sys`), and proves the system has only one solution (`SHH.Walk.sys_unique`); together these make the closed form the expected number of steps (`SHH.Walk.sys_eq_s`). The paper's Eq. 2 as printed,

$$s = \frac{2}{(2p_- - 1)^2}\left[p_-^2 + p_-(D-1) - \frac{D}{2} + \frac{(1-p_-)^{D+1}}{p_-^{D-1}}\right],$$

is exactly this function for every p_- other than 0 and $\frac{1}{2}$ (`SHH.Walk.eq2_printed`). The 1988 derivation was right.

One detail had to be decided before any of this could be stated: what the walk does at distance D, where the paper says only that it is "constrained to remain within distance D". The paper's own value for the symmetric case, $s = D^2$, settles it: that value holds if the walk is reflected back at D (`SHH.Walk.s_half`), and not if it may stay there, which gives $D^2 + D$ (`SHH.Walk.stay_half_sys`). A choice the paper left implicit is recoverable from what it reported.

The regimes. When steps toward the goal are more likely than steps away ($p_- > \frac{1}{2}$), the paper's Eq. 3 says the walk drifts to the goal in about $D/(p_- - p_+)$ steps. This holds as a two-sided bound: the expected steps lie within $2p_-(1 - p_-)/(2p_- - 1)^2$ below $D/(2p_- - 1)$, a constant that does not grow with D (`SHH.Walk.drift_bounds`). When steps away are more likely, the paper's Eq. 4 gives the exponential term of Eq. 2, and the rest is a term only linear in D (`SHH.Walk.eq4_remainder`, `SHH.Walk.s_diffusion_form`). In every regime, more distance costs strictly more steps (`SHH.Walk.s_strictMono_D`).

4 What the theory entails, and what it does not

The paper then turned the walk into claims about practice. Learning adds edges, so μ grows; more edges shorten the diameter, which helps, and add bad choices, which hurts. Two of the paper's conclusions about which effect wins do not follow.

A fan effect where the paper says there is none. For a weak decision procedure the paper approximates Eq. 4 by its Eq. 5, $s \approx (2n/\mu)(1 - r)^{\ln n / \ln \mu - 1}$, and concludes that "when $r = 0$ (choices made at random), increasing the number of edges reduces the time to solve the problem". The approximation replaces $(\mu - 1)^{D-1}$ by $\mu^{D-1} = n/\mu$, dropping a factor $((\mu - 1)/\mu)^{D-1}$ that is at most $e^{-(D-1)/\mu}$ (`SHH.GD.eq5_error_bound`). The paper states Eq. 5 for $\mu(1 - r) \gg 1$, and there the

factor is harmless. But the conclusion about $r = 0$ is stated without that proviso, and in a sparse graph, where μ is small, the factor is far from one. Restoring it reverses the conclusion there. With $n = 2^{16}$ nodes, where the diameter is an integer for $\mu = 2, 4, 16, 256$ (`SHH.GD.DOf_pow2`), the paper's own idealized model gives the expected steps in Table 1: at $r = 0$, raising the mean branching from 2 to 4 raises the expected steps from 256 to 9,824, and by $\mu = 16$ they have come back down only to 7,744 (`SHH.GD.fan_pow2_zero`; numerically the peak is near $\mu = 6$). In the theory, in a sparse enough graph, there is a fan effect with no decision procedure at all. The same happens at the $n = 10{,}000$ of the paper's Figure 2, where the prototype has the expected steps rising from $\mu = 2$ until about $\mu = 5$. Whether any of this touches the bit game is a separate question, and Section 5 takes it up: the game is never that sparse.

μ	D	$r = 0$	$r = \frac{1}{2}$	$r = 1$
2	16	256	44.0	16
4	8	9,824	255.5	8
16	4	7,744	1,117.75	4
256	2	512	257	2

Table 1: Expected steps in the paper's idealized model, $n = 2^{16}$, from the proved closed form (`SHH.GD.fan_pow2_zero`, `SHH.GD.fan_pow2_half_values`, `SHH.Walk.s_one`). The $r = \frac{1}{2}$ entries are rounded; the exact values are in the theorem.

A logarithm where the paper says power law. With a strong decision procedure the paper's Eq. 6 gives $s \approx \ln n / \ln \mu$ and concludes that "one obtains a power-law decay". At $r = 1$ the approximation is exact: the walk never steps away, so $s = D$ (`SHH.Walk.s_one`). To read this as a learning curve one needs a further condition the paper leaves implicit, that each trial adds a bounded number of edges, so that μ grows about linearly with trials. The curve is then $T(t) = \ln n / \ln(\mu_0 + ct)$ (`SHH.GD.curveGD`). It improves with every trial (`SHH.GD.curveGD_improves`), and its proportional gains shrink trial by trial (`SHH.GD.curveGD_diminishingRelative`), the signature that distinguishes a power curve from an exponential one [16]. Over trials 1–100, with $n = 10^4$, $\mu_0 = 2$ and $c = 1$, it lies within about 12% of a power law with exponent -0.29 (`tools/walk_check.py`). But it is not a power law. A power law shrinks by a fixed ratio $\rho < 1$ every time the number of trials doubles; this curve's doubling ratio $T(2t)/T(t)$ tends to 1, so for any such ρ, and any tolerance too small to reach 1 from it, approximate scale invariance fails beyond some trial (`SHH.GD.curveGD_not_powerLaw`). The curve is logarithmic. On a log-log plot over the one decade of Figure 1 of the 1988 paper, the difference is invisible.

This does not undo the 1988 argument against building the power law into the mechanism. It sharpens it. The graph-dynamic model does not build in a power law, and, at least in its perfect-decision regime, it does not produce one either. What it produces is a curve that cannot be told apart from one over the range anyone plotted.

5 The game itself

The theory's conclusions are about an idealization. Does the bit game itself behave as the theory says? The formalization can answer this exactly for small games, because the expected number of steps in the game is the solution of a finite linear system, and that solution is unique whenever the goal is reachable (`SHH.Markov.hit_unique`, proved by the maximum principle). So a table of values checked to satisfy the system is the expected number of steps.

On the three-bit game, adding one learned operator lowers the mean expected steps over all 56 games at $r = 0$ and at $r = 1$, and raises it at $r = \frac{1}{2}$ (Table 2; `SHH.BitGame.figure1_pattern_three`). The prototype finds the same for every one of the 20 new operators that could be learned. This is the pattern of the 1988 simulations: improvement with no decision procedure and with a perfect one, degradation with a mediocre one. (Individual games can still slow down at $r = 0$: `000` $\rightarrow$ `001` goes from 7 to 133/15 expected steps, `SHH.BitGame.chunk_slows_game`. It is the average that improves.)

How does this compare with the theory? The comparison has to be made where the game is. A B-bit game has 2^B states and starts as a hypercube, with branching $\mu = B$, and learning only adds edges from there. Table 1 has 2^{16} nodes, the size of a sixteen-bit game, but its first two rows, $\mu = 2$ and 4, are graphs far sparser than that game ever is. Read from $\mu = 16$ down, the $r = 0$ column only improves, 7,744 to 512. The same holds at four bits (`SHH.GD.hypercube_start_falls`), and in the prototype at every size I have tried from 3 to 24 bits: the rise at $r = 0$ always lies below the hypercube's own branching. So for the bit game the 1988 sentence was right. At $r = 0$ more edges do reduce the time, and theory and game agree. What Section 4 refutes is the general statement, not its application to the game.

At $r = \frac{1}{2}$ it is the other way round. The game has a fan effect: the simulations showed it at five bits, and Table 2 shows it at three. The theory, started at $\mu = B$, does not. At four bits the expected steps go from 5 to 1, and at sixteen from 1,117.75 to 257 (the same theorem); in the prototype they fall steadily at five bits, and at every size up to about twenty bits, where a rise of about one percent first appears. The 1988 Figure 2 plotted Eq. 2 for 10,000 nodes, where a fan effect at $r = \frac{1}{2}$ does appear in sparse graphs, and found the simulations' pattern in it. At the 32 states and the branching of the simulations themselves, the same equation predicts steady improvement in every regime. The fan effect in the bit game is real, but the walk is not what explains it. I do not yet know what does. The comparison is rough, since a tree with a diameter of two or three is a poor picture of anything, and one learned operator is not a uniform increase in branching; but that is a reason to doubt that the theory applies at this size, not a reason to think it explains what happens there.

	$r = 0$	$r = \frac{1}{2}$	$r = 1$
initial operators	$58/7 \approx 8.286$	$128/35 \approx 3.657$	$12/7 \approx 1.714$
plus the chunk `00?` $\rightarrow$ `11?`	$2603/315 \approx 8.263$	≈ 3.682	$23/14 \approx 1.643$

Table 2: Mean expected steps over the 56 games of the three-bit game, exact (`SHH.BitGame.mean_r0_Base` and its five companions).

Two results about the perfect decision procedure complete the picture. With the initial operators, every move it can make brings the state one bit closer to the goal, for any number of bits, so it never searches or backtracks (`SHH.BitGame.r1_moves_closer`). Once operators have been learned, this fails: on the four-bit game, with the learned operator `0000` $\rightarrow$ `1111`, the game `0001` $\rightarrow$ `1110` has a three-move solution (`SHH.BitGame.three_step_path`) but the perfect procedure expects 15/4 moves (`SHH.BitGame.greedy_not_optimal`). Choosing the move that looks best by distance can miss the shortcut that learning created. The 1988 paper said as much, that its $r = 1$ procedure "only approximates optimal problem solving"; here is a case, exactly.

Claim of the 1988 paper	Status
Eq. 2, the expected steps	proved, exactly as printed (`SHH.Walk.eq2_printed`)
$s = D^2$ when $p_+ = p_-$	proved; fixes reflection at D (`SHH.Walk.s_half`)
Eq. 3, drift	proved as a two-sided bound (`SHH.Walk.drift_bounds`)
Eq. 4, diffusion	proved as Eq. 2's exponential term (`SHH.Walk.eq4_remainder`)
Eq. 5 and "at $r = 0$ more edges reduce the time"	refuted as a general statement, in sparse graphs (`SHH.GD.fan_pow2_zero`); true from the bit game's starting point (`SHH.GD.hypercube_start_falls`)
Fan effect at intermediate r	proved in the theory for sparse graphs (`SHH.GD.fan_pow2_half`) and in the three-bit game (`SHH.BitGame.figure1_pattern_three`)
Eq. 6 and "a power-law decay"	reinterpreted: logarithmic, not a power law (`SHH.GD.curveGD_not_powerLaw`)
Eq. 2 "produces the range of behaviors observed"	not from the game's starting point: there the theory has no fan effect at $r = \frac{1}{2}$, and the game does (`SHH.GD.hypercube_start_falls`, `SHH.BitGame.figure1_pattern_three`)
$r = 1$ approximates optimal search	proved optimal on the initial operators (`SHH.BitGame.r1_moves_closer`); not optimal with learned ones (`SHH.BitGame.greedy_not_optimal`)

Table 3: The 1988 claims and their status. The full list is in `notes/CLAIMS.md`.

6 Discussion

The 1988 paper was mathematically careful, and most of its mathematics survives a machine check unchanged. What did not survive were two inferences from the mathematics to the psychology: an approximation applied where it is poor, and a curve called by the name of the curve it resembles. Neither is an unusual kind of mistake. Both are the kind a proof assistant catches almost incidentally, because it will not accept "approximately" without saying approximately what, over which range, and why.

The more interesting finding is about idealization. The 1988 paper moved from the bit game to a random graph to a tree to a walk in a few sentences, and treated the walk's behavior as the game's. In the formalization the game and the walk are separate objects, and the step between them is a postulate with a name. That is what made it possible to ask whether a conclusion of the theory holds of the game, and to find that the theory's fan effect and the game's are not the same effect: one needs a graph sparser than the game ever is, and the other was observed in a hypercube of 32 states. I think this is the general lesson. Idealizations are where theories make their least visible commitments; writing them down as first-class objects, rather than as a sentence of motivation, makes them something one can test.

This is also why the work belongs in a shared library rather than a single paper. The phenomena it uses, a curve that improves with diminishing relative returns and a curve that worsens, are stated without reference to the graph-dynamic model, so the same predicates can be put to ACT-R's base-level learning or to chunking accounts and the answers compared, as the CogLean project (`https://coglean.org`) does for theories of memory [11].

It also says something about the 1988 argument with ACT*. That argument was that a theory should not get the power law of practice by building it into a mechanism. The graph-dynamic model meets that standard, but the curve it yields in its cleanest regime is logarithmic, and the

evidence for the power law has itself been contested [16]. The defensible claim now is narrower and, I think, more interesting: that improvement with diminishing relative returns, the shape that separates a power law from an exponential and the very thing Heathcote and colleagues dispute in the data of individual learners, emerges from search dynamics without any learning curve being assumed, and that the same dynamics, with a mediocre decision procedure, make practice hurt.

Limitations. The game results are for three and four bits, and for one learning step at a time; the 1988 simulations used five bits and a hundred trials of learning. Reproducing Figure 1 of that paper by simulation, with the sampler checked against the exact small cases, is the next step, and it will show whether the game's fan effect at five bits behaves as it does at three. The tree idealization is stated, not derived from random-graph theory [17], and the exact results in the theory use settings where the diameter is an integer, which for a game started as a hypercube means four bits and sixteen. At three and five bits, the sizes of the exact game results and of the 1988 simulations, the comparison with the theory rests on the prototype's numbers. Nothing here concerns people: the phenomena are properties of the model's behavior, and connecting them to human practice data needs the usual assumptions about what is measured.

7 Conclusion

The 1988 paper argued that learning need not improve performance, and that whether it does depends on how decisions are made. Revisiting it with a proof assistant confirms that argument, corrects two of its supporting claims, and reopens a question the paper thought it had closed: why the bit game shows a fan effect at all. All of this came from writing the theory down, not from new mathematics. A model that only runs can show that a behavior occurs; a model one can reason about shows what it entails, where its approximations hold, and which of its idealizations is responsible for a result. Thirty-eight years on, the graph-dynamic model has more to say than its authors knew.

Availability

All definitions and theorems are in SHH88Revisited [14] (Lean 4.35.0, Mathlib v4.35.0). Every theorem is fully proved, with no `sorry`, and depends only on Lean's standard axioms. The Python prototypes in `tools/` have the same semantics as the Lean definitions and generate the exact certificates.

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